

SECTION 'A'

1) 4 $\Rightarrow$ Argon

2) 2 $\Rightarrow$ calcium

3) 4 $\Rightarrow$ 22

4) 1 $\Rightarrow$ 17

5) 3 $\Rightarrow$ 1:1

6) 4 $\Rightarrow$ Mercury

7) 2 $\Rightarrow$ Lead

8) 1 $\Rightarrow$ 1

9) 1 $\Rightarrow$ Lithium

10) 4 $\Rightarrow$ Ammonium chloride

11) 4 $\Rightarrow$ Copper chloride

12) 2 $\Rightarrow$ consists of molecules

13) 3 $\Rightarrow$ An alkali

14) 1 $\Rightarrow$ Direct combination

15) 3 $\Rightarrow$ conc. HCl + conc. HNO_3

16i) a) $\Rightarrow$ Z

b) L

c) Halogen Group

d) Nitrogen of Group 15 and Boron of group

e) O has smaller radius

ii a) reducing agents

b) bad

c) acidic

d) Silver Chloride

e) Cl

$$\begin{aligned}\text{iii) i) Molecular wt of } \text{SO}_2 &= (32) + (2 \times 16) \\ &= 32 + 32 \\ &= 64 \text{ amu}\end{aligned}$$

$$\text{No. of moles} = \frac{\text{wt}}{\text{molecular wt}}$$

$$= \frac{320}{64} \quad 5$$

$\Rightarrow$ 5 moles

1 mol contains 22.4 l at STP

$$\begin{aligned}\therefore 5 \text{ mol will contain} &= 22.4 \times 5 \text{ l} \\ &= 112.0 \text{ l}\end{aligned}$$

112.0 l

ii) According to it when gases react, they do so in volumes which bear a simple ratio to one another, and to the volume of the gaseous product, provided that all the volumes are measured at the same temperature and pressure.

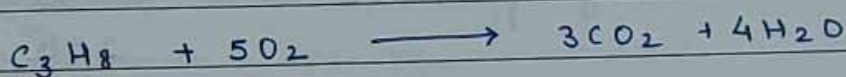
iii) mass of propane = 8.8 g Molar mass of C_3H_8

$$= (12 \times 3) + (8 \times 1)$$

$$= 36 + 8 = 44$$

$$\text{No. of moles} = \frac{\text{mass}}{\text{molar mass}}$$

$$= \frac{8.8 \text{ g}}{44 \times 10^{-3}} = \frac{1}{5} = 0.2 \text{ mol}$$



1 mol of propane requires $5 \times 22.4 \text{ l}$ of O_2

$\therefore$ 0.2 mol will require $5 \times 22.4 \times 0.2$

$$= \frac{5 \times 22.4}{100} \times 20$$

$$= \underline{\underline{22.4 \text{ l of } O_2}}$$

$$\text{iv i) No. of gram atoms} = \frac{\text{wt}}{\text{molecular wt}}$$

$$= \frac{4.6 \text{ g}}{28 \times 10} = \frac{1}{5}$$

$$= \underline{\underline{0.2 \text{ g atoms}}}$$

$$\text{ii) Molecular wt of water} = 18 \text{ g}$$

$$\begin{aligned} \text{Molecular wt of } \text{CuSO}_4 &= 64 + 32 + (4 \times 16) + (5 \times 18) \\ &= 250 \text{ g} \end{aligned}$$

$$\text{Percentage of water} = \frac{5 \times 18 \text{ g} \times 100}{250}$$

$$= \underline{\underline{36\%}}$$

$$\text{iii) Empirical formula} = \text{XY}_2$$

$$\text{Molecular formula} = 2 \times \text{XY}_2$$

$$= \underline{\underline{\text{X}_2\text{Y}_4}}$$

$$\text{vi) } 15$$

$$\text{ii) } 19$$

$$\text{iii) } 4$$

$$\text{iv) } 2$$

SECTION B

$$\begin{aligned} \text{i) Molar mass of } C_3H_4N &= 3 \times 12 + 4 \times 1 + 1 \times 14 \\ &= 36 + 4 + 14 \\ &= 54 \text{ g/mol} \end{aligned}$$

$$\begin{aligned} \text{Molecular formula} &= 2 \times \text{Empirical formula} \\ &= 2 \times C_3H_4N \\ &= C_6H_8N_2 \end{aligned}$$

$$\text{No. of carbon} = \underline{\underline{6}}$$

ii) It will become less i.e. $\text{pH} < 7.3$ or acidic

ii) It will become alkaline and pH will increase

b) 15th Group



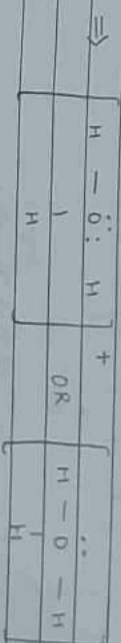
$$\begin{array}{l} 164 \text{ g} \\ 2 \times 164 \\ \Rightarrow 328 \text{ g} \end{array} \quad \begin{array}{l} 2 \times (40 + 16) \\ 4 \times 22.4 \text{ l} \end{array}$$

$$\begin{aligned} \text{Moles of Ca(NO}_3)_2 &= \frac{8.2}{164} \times 164 \\ &= 8.2 \times 10^{-2} \\ &= 0.05 \text{ moles} \end{aligned}$$

$$\begin{aligned} \text{Moles of NO}_2 &= 2 \text{ moles of Ca(NO}_3)_2 \\ &= 0.100 \text{ moles} \end{aligned}$$

22.4 l of NO_2 will be obtained

iii)



iv) i) When NaOH reacts with zinc it forms a soluble complex compound of sodium zincate which is a colourless solution while with aluminium it does not react.

ii) When ammonium hydroxide is added to calcium nitrate solution a white ppt of calcium hydroxide is formed while with a solution of lead nitrate it forms a brown ppt of lead (II) hydroxide.

2) i) Structure: Cations are charged particles which are formed by losing electrons. This reduces their electron count to match its nearest noble gas configuration.

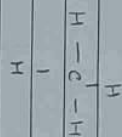
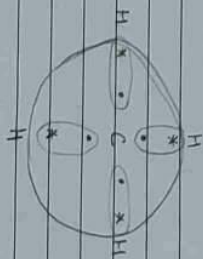
ii) The compound Pb_2 is an ionic compound. In molten state, Pb_2 an ionic compound's rigid band is broken and the ions become mobile (and in aqueous state too) thus it becomes a good conductor of electricity.

ii) No. of moles = $\frac{\text{mass}}{\text{molar mass}}$

$$= \frac{22}{44} \times \frac{1}{2} = 0.5 \text{ moles}$$

- iii) a) Oxidising power: $X > Y$
 b) Electro negativity: $X > Y$
 c) X is to the right of Y

iv-i)



4 single covalent bonds -

ii) a) Barium will form ions most readily due to its large size and low ionization energy.

b) They have tendency to remove electron

3 f) $M.O. =$ Since valency of O is -2 $\therefore$ M will be +2.

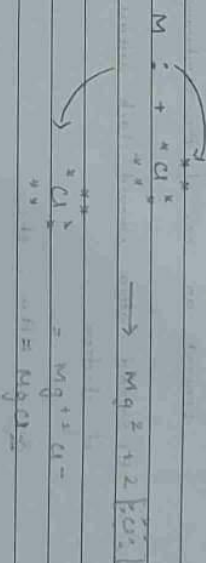
Sulphur = MS

Chlorine = MCl_2

ii) a) Hydrogen and oxygen form single covalent bonds because it has only one electron to share to achieve a stable configuration while oxygen forms double covalent bonds as it has 6 valence electrons and it needs 2 more to achieve a stable configuration of 8 electrons.

b) In CCl_4 carbon shares one electron with each chlorine to complete its octet and chlorine atoms receive one electron each to become stable. Thus, CCl_4 forms 4 single covalent bonds with each atom.

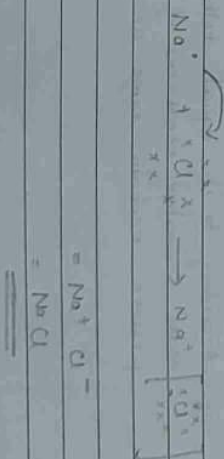
iii) i) $MgCl_2$



ii) $CaO \Rightarrow$



iii) $NaCl$



iv) Coordinate bond is formed by sharing of an electron pair from a single atom.

3 conditions are :

a) At least one lone pair of electron should be present on one of the combining atoms.

b) Second atom should have at least one pair of electron.

Eg : NH_4^+ , etc.

4i) $X = 2, 8, 8, 2$

ii) Period = 4 Group 2

ii) Valency = +2

b) Ionic compound

iii) According to it equal volumes of all gases under similar conditions of temperature and pressure contain same no. of molecules.

Some applications are :

a) Determine the atomicity of gases

b) Determination of molecular formula of gaseous compound

c) Establishment of relationship between gram molecular weight and volume of gas at STP.

iv-i) A and C

ii) A and C

iii) B and D

5i) $\text{CaCO}_3 \longrightarrow \text{CaO} + \text{CO}_2$

100g 56g

1 mol of CaO obtained by 1 mol of CaCO_3

56g of CaO will be obtained from 100g of CaCO_3

$\therefore 14\text{g}$ residue will be formed from $\frac{14 \times 100}{56}$

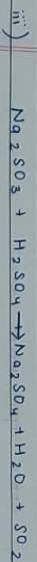
25g of CaCO_3

ii) $2\text{C} + \text{O}_2 \longrightarrow 2\text{CO}_2$

20g 20g

1 molecule = CO_2

$\therefore$ 10 molecules of CO_2 will be formed and 10g C will be left.



$$\text{Molar mass of SO}_2 = 32 + 2 \times 16 = 64 \text{ g}$$

$$\text{Molar mass of Na}_2\text{SO}_3 = 2 \times 23 + 32 + 16 \times 3 = 124 \text{ g}$$

$$\text{Moles of SO}_2 = \frac{2 \times 1}{64 \times 10} = \frac{1}{20} = 0.05 \text{ moles}$$

$$\begin{aligned} \text{Mass of Na}_2\text{SO}_3 &= 0.05 \times 124 \\ &= 6.20 \\ &= 6.20 \text{ g of Na}_2\text{SO}_3 \text{ is required} \end{aligned}$$

iv-i) copper and aluminium hydroxides.

ii) Sodium sulphate (Na_2SO_4) is an acidic salt and Sodium bisulphate (NaHSO_4)

iii) lead chloride

6 i) a) Blue vitriol ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$)

b) Sodium chloride (NaCl)

ii) Deliquescence is the process by which a substance absorbs moisture from atmosphere and dissolves in it when exposed in open and forms a solution.

Effluence is the property of substance to lose its water of crystallisation when exposed to dry air.

iii) Properties of acid are

a) Acid reacts with metals to form salts and releases H_2 gas.

b) Acids react with bases to form salt and water

c) Acids have a pH value less than 7

iv) $\text{H}_3\text{PO}_4 = 3$

b) $\text{Co(OH)}_2 = 2$

c) $\text{NaOH} = 1$

d) $\text{CH}_3\text{COOH} = 1$

e) $\text{H}_2\text{SO}_3 = 2$

f) $\text{H}_3\text{PO}_3 = 2$